Summary in Graph

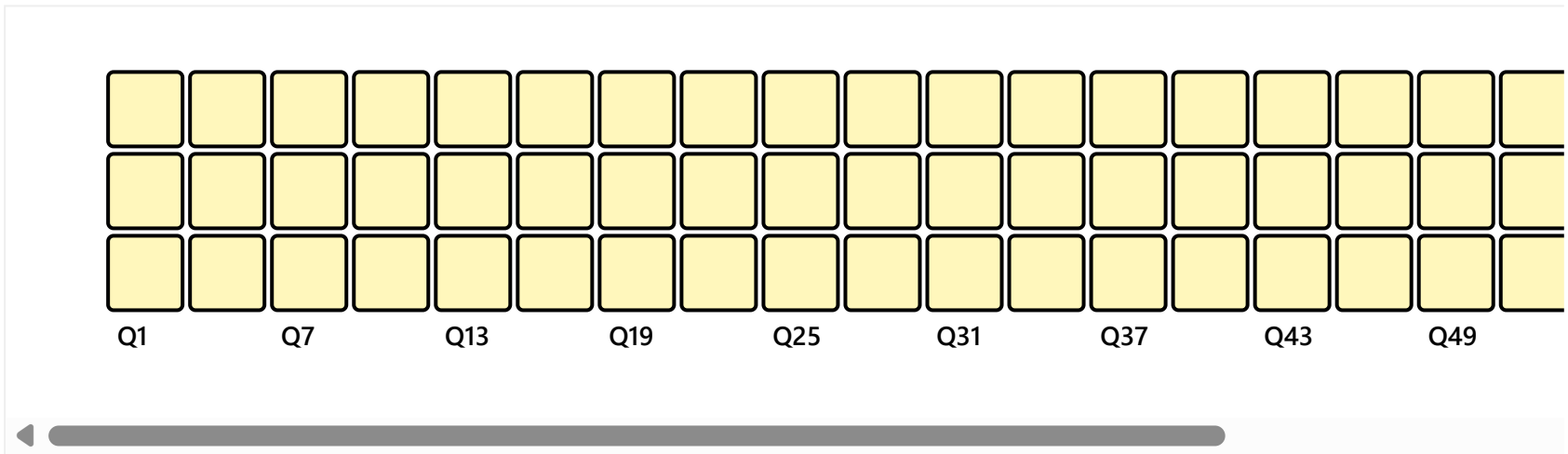
Exam Summary (GO Classes CS Test Series 2026 | Mixed Subjects | Full Length Test 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (36) out of 359 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose numbers from 1 to 1000 are saved in a binary search tree and we want to find 363. Which of the following sequences cannot be the order of elements while reaching the searched value?

A. $925, 202, 911, 240, 912, 245, 363$
B. $924, 220, 911, 244, 898, 258, 362, 363$
C. $2, 252, 401, 398, 330, 344, 397, 363$

D. $2, 399, 387, 219, 266, 382, 381, 278, 363$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

You are given a minimum heap with 31 elements. A researcher performs a uniform walk on this heap. Given any node, with probability 0.5 she goes to the left. Starting from the root node, what is the probability that she will land up on a maximum element in the heap? Assume all elements are distinct.

A. $1/31$
B. $1/32$
C. $1/16$
D. $1/15$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Assume that any N-bit positive integer x is stored as a linked list of bits so that the first element of the list is the least significant bit. For example, $x = 14 = 1110_2$ is stored as the linked list $(0, 1, 1, 1)$ of size $\mathrm{N} = 4$. For this data structure, the operation that replaces x by $\left\lfloor \frac{x}{8} \right\rfloor$ can be done in

A. $\Theta(1)$ steps
B. $\Theta(\log n)$ steps
C. $\Theta(n)$ steps
D. $\Theta(n \log n)$ steps

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

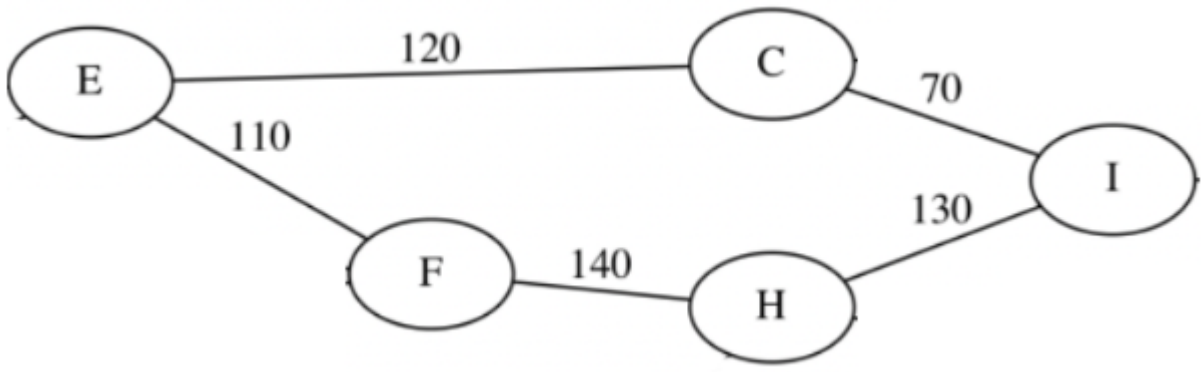
For which of the following functions $f(n)$ and $g(n)$, it holds : $f(n) = O(g(n))$. Every $\log$ below is base 2.

A. $f(n) = 2^{k \log n}\ , \quad g(n) = n^k$
B. $f(n) = 2^n\ , \quad g(n) = 2^{2n}$
C. $f(n) = \begin{cases} 4^n & \text{if } n < 2^{1000} \\ 2^{1000}\ n^2 & \text{if } n \geq 2^{1000} \end{cases}\ , \quad g(n) = \dfrac{n^2}{2^{1000}}$
D. $f(n) = 2^{\sqrt{\log n}}\ , \quad g(n) = (\log n)^{100}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the following connected weighted graph.



Suppose we add a new edge $\mathrm{H}-\mathrm{C}$ to the above graph.

For what values of $\mathrm{H}-\mathrm{C}$ does this new edge belong to the Minimum Spanning Tree (MST)?

A. 125
B. 100
C. 135
D. 90

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Let N be an arbitrary natural number with $\mathrm{N}>1$. Which one of the following statements is false?

A. A binary tree of height N may have $\mathrm{N}+1$ nodes.
B. A binary tree of height N may have N^2 nodes.
C. A binary tree of height N may have 2^{N} nodes.
D. A binary tree of height N may have $\mathrm{N}!$ nodes.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Assume a set A of arrays of size N all of which are sorted already. Which one of the following statement is correct assuming there are 10 such arrays in the set?

A. Mergesort on elements of A may take a runtime in $\Omega\left(\mathrm{N}^2\right)$.
B. Insertion Sort on elements of A may take a runtime in $\Omega\left(\mathrm{N}^2\right)$.
C. Heapsort on elements of A may take a runtime in $\Omega\left(\mathrm{N}^2\right)$.
D. None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which statement on Mergesort is correct?

A. There are arrays of size N such that Mergesort runs in $\Theta\left(\mathrm{N}^2\right)$.
B. There are arrays of size N such that Mergesort runs in $o\left(\mathrm{N}^2\right)$.
C. There are arrays of size N such that Mergesort runs in $O(\mathrm{N})$.

D. There are arrays of size N such that Mergesort runs in $\Omega\left(\mathrm{N}^2\right)$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Some sorting algorithms require extra space, apart from the space needed for the original array that needs to be sorted. Which one of the following statements on the space usage of sorting algorithms is correct?

A. Heapsort for sorting an array of size N requires an amount of extra space proportional to N.
B. Insertion Sort for sorting an array of size N requires an amount of extra space proportional to N.
C. Mergesort for sorting an array of size N requires an amount of extra space proportional to N.
D. Quicksort for sorting an array of size N requires an amount of extra space proportional to N.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Assume that you are given a chain of matrices $< \mathrm{A}_1\mathrm{A}_2\mathrm{A}_3\mathrm{A}_4 >$, with dimensions $2 \times 5, 5 \times 4, 4 \times 2$ and 2×4 respectively. Compute the optimal number of multiplications required to calculate the chain product.

Your Answer: | Correct Answer: 72 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the given two statements.

- $\mathrm{S1}$: Depth-first search is asymptotically faster than breadth-first search.
- $\mathrm{S2}$: Deleting an element from a binary search tree takes $\mathrm{O}(\lg n)$ time in the worst case.

Which of the following is correct about $\mathrm{S1}$ and $\mathrm{S2}$?

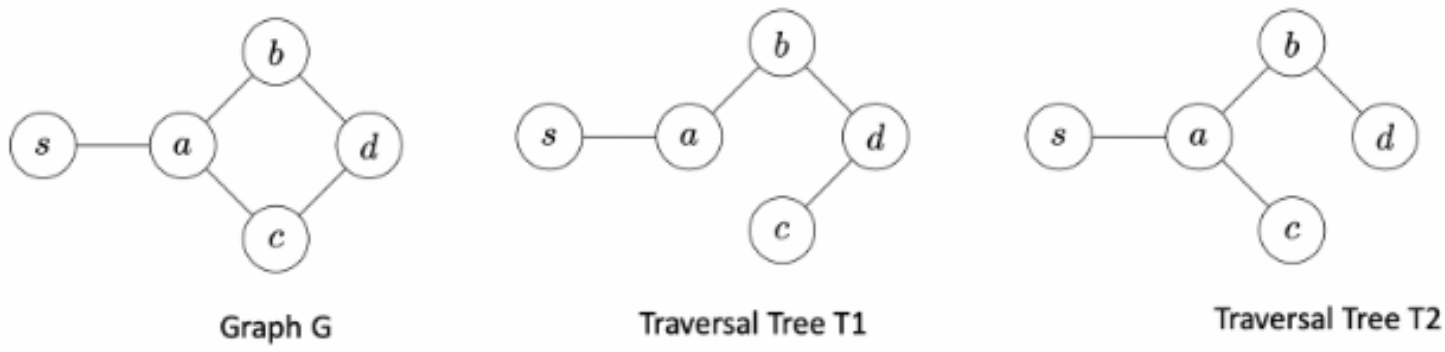
A. $\mathrm{S1}$ is correct but $\mathrm{S2}$ is wrong.
B. $\mathrm{S2}$ is correct but $\mathrm{S1}$ is wrong.
C. Both are correct
D. Both are False

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the following graph G.

Traversal trees $T1$ and $T2$ (shown below) are obtained using either DFS or BFS, starting from the node S.



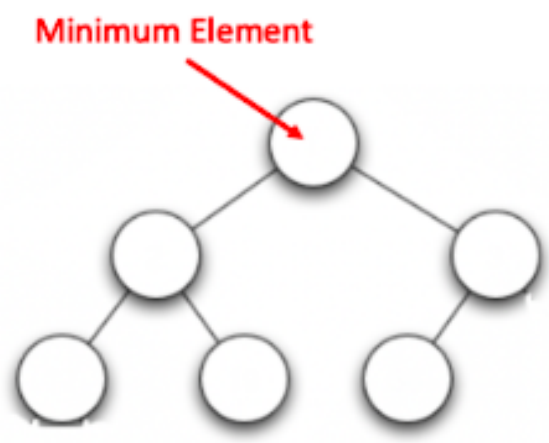
Which of the following statements is/are correct?

A. $T1$ is possible only using DFS on G
B. $T1$ is possible using DFS or BFS on G
C. $T2$ is possible only using BFS on G
D. $T2$ is possible using DFS or BFS on G

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a min-heap with 6 distinct elements. How many positions can be taken by 3rd minimum element?



A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following are greedy algorithms?

A. Prim's algorithm for MST.
B. Dijkstra's algorithm for shortest paths.
C. Bellman-Ford algorithm for shortest paths.
D. Finding the longest common subsequence of two sequences.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following is/are TRUE ?

A. If we perform DFS on an undirected graph, there are no cross edges.
B. If the DFS tree has no back edges, then there are no cycles in the graph.
C. Dijkstra's algorithm will always work correctly on any graph that has at most two negative edges.
D. Dijkstra's algorithm will always work correctly on any graph with negative edges but no negative cycle.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Suppose we want to iteratively evaluate the following recurrence:

$$\text{What }(i,j) = \begin{cases} 0 & \text{if } i > n \text{ or } j < 0 \\ \max \left\{ \begin{array}{c} \text{What }(i, j-1) \\ \text{What }(i+1, j) \\ A[i] \cdot A[j] + \text{ What }(i+1, j-1) \end{array} \right\} & \text{otherwise} \end{cases}$$

Which of the following is/are true about filling table What $[0 \ldots n, 0 \ldots n]$ in a bottom-up fashion?

A. We can fill the table in $O\left(n^2\right)$ time
B. We can fill the table by decreasing i in the outer loop and decreasing j in the inner loop.
C. We can fill the table by increasing i in the outer loop and increasing j in the inner loop.
D. We can fill the table by decreasing i in the outer loop and increasing j in the inner loop.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following C program given below.

```c
#include<stdio.h>
main() {
    int a = 4;
    switch (a) {
    a--;
    case 4:
        printf("Science ");
        break;
    default:
        printf("Technology ");
    case 3:
        printf("Knowledge ");
    case 2:
        printf("Philosophy");
    }
}
```

What will be the output of the program?

A. Science
B. Knowledge Philosophy
C. Technology Knowledge Philosophy
D. Science Knowledge

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type Award: 1 Penalty: 0 Programming in C

Consider the following declaration of pointer variable p.

```
int (*p)[10][5];
```

If the initial value of p is 1000, then what will be the value of $p+1$?

It is given that system has 8 bytes of address size and 4 bytes of integer size.

Your Answer: Correct Answer: 1200 Not Attempted Time taken: 00min 00sec Discuss

Q #19 Multiple Select Type Award: 1 Penalty: 0 Programming in C

Let arrays OneD and TwoD are declared as follows as:

```
int OneD[10];
int TwoD[4][5];
```

Which of the following is/are valid syntax to pass OneD and TwoD to some function fun()?

A syntax is valid if it gets compiled without any warning or error.

A.
```
void fun(int OneD[10]);
```

B.
```
void fun(int OneD[5]);
```

C.
```
void fun(int TwoD[][5]);
```

D.
```
void fun(int **TwoD);
```

Your Answer: Correct Answer: A;B;C Not Attempted Time taken: 00min 00sec Discuss

Q #20 Multiple Choice Type Award: 1 Penalty: 0.33 Programming in C

```
#include <stdio.h>
int main()
{
    int i= 255;
5.  short int *s= (short int *)&i;
    printf("%d\n", *s);
}
```

What will be the output of the above program in little-endian and big-endian, respectively?

$(65280 \text{ is } 255 \times 2^8)$

A. $255,\ 0$
B. $65280,\ 0$
C. $0,\ 0$
D. $0,\ 65280$

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #21 Numerical Type Award: 1 Penalty: 0 Programming in C

What will be the number of recursive calls for mystery(5) including the first call?

```
void mystery(int n) {
    if (n == 0 || n == 1) return 0;
    mystery(n-2);
    printf("%d", n);
    mystery(n-1);
}
```

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

In which of the following case(s) character array must end with null char?

A.
```
char c[] = "GATE";
```

B.
```
char c[] = {'2', '0', '2', '3'};
```

C.
```
char c[4] = "GATE";
```

D.
```
char c[16] = "2023";
```

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Consider the following C code fragement -

```
#include<stdio.h>
int main()
{
    char t[] = "abcdefghij";
    int *p = t;
    p+=2;
    char *c = (char *)p;
    printf("%c", *c);
}
```

Which of the following is/are correct?

A. The output of the program in the little-endian system is " i " (without quotes).
B. The output of the program in the big-endian system is " i " (without quotes).
C. The output of the program in the big-endian system is " c " (without quotes).
D. The output of the program in the little-endian system is " c " (without quotes).

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Assume that an int variable takes 4 bytes and a char variable takes 1 byte. What is the output of the code below?

```
int main()
{
    int arr[]={10,20,30,40,50,60};
    int *ptr1=arr;
    int *ptr2=arr+5;
    printf("Number of elements between two pointer are: %d.",
    (ptr2 - ptr1));
    printf("Number of elements between two pointer are: %d.",
    (char*)ptr2 - (char*)ptr1);
}
```

A. Number of elements between two pointer are: 5. Number of bytes between two pointers are: 20
B. Number of elements between two pointer are: 20. Number of bytes between two pointers are: 20
C. Number of elements between two pointer are: 5. Number of bytes between two pointers are: 5
D. Compile time error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Which of the following function declaration can be passed to the following array?

```
char myArray[3][4];
```

A.
```
void fun(char a[][], int size);
```

B.
```
void fun(char a[][4], int size);
```

C.
```
void fun(char [3][], int size);
```

D.
```
void fun(char [][]a, int size);
```

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Heaps are usually implemented using arrays. Which of the following statements is correct about the removal of the element?
The removal of an element with known array index from an array of size N requires a time of- (array index could be arbitrary. Not necessarily delete-min)

A. $\Theta(\log \mathrm{N})$
B. $\Theta(\mathrm{N})$
C. $\Theta(1)$
D. $\Theta\left(\mathrm{N}^2\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

An algorithm admits all 2^n vectors of n bits as possible inputs. On $2^n - 1$ of these inputs the cost of the algorithm is $\Theta\left(n^2\right)$, and on the remaining one the cost is $\Theta\left(n^4\right)$. Therefore, its worst-case cost is $\Theta\left(n^4\right)$ and its best-case cost is $\Theta\left(n^2\right)$. What is its average-case cost when the input is chosen uniformly at random (so each input has probability $1/2^n$)?

A. $\theta\left(n^2\right)$
B. $\theta\left(\mathrm{n}^3\right)$
C. $\theta\left(\mathrm{n}^4\right)$
D. $\theta\left(2^n\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

What is the sum of the following numbers?

$$50+53+56+59+\ldots+3080+3083+3086=?$$

Your Answer: | Correct Answer: 1588384 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

To print all keys of a BST in decreasing order, we use the following traversal order

A. Preorder traversal
B. Postorder traversal
C. Inorder traversal
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose you have a five element integer array $\mathrm{A}[0 \ldots 4]$ sorted in non-decreasing order, with $\mathrm{A}[0]=1$ and $\mathrm{A}[4]=5$. We search for key $\mathrm{K}=2$.

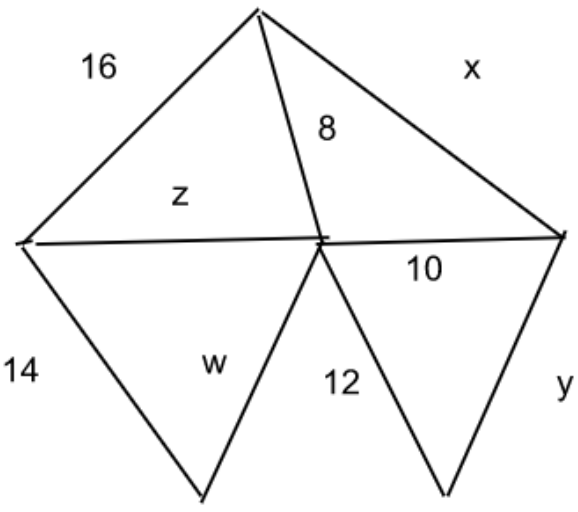
How many possible set of values are there for elements $\mathrm{A}[1], \mathrm{A}[2], \mathrm{A}[3]$ such that binary search terminates immediately after one comparison?

A. 1
B. 2
C. 4
D. 8

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

The weight of the unique minimum weight spanning tree (MWST) is 60. The weight of edges which forms MWST is given in the following weighted, undirected graph. What will be the minimum total weight of missing edges, i.e., $w+x+y+z=?$. Assume all edge weights are distinct.



Your Answer: | Correct Answer: 59 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider a Job scheduling problem with deadlines.

Jobs	1	2	3	4	5	6	7
Profit	8	6	5	X	8	3	4
Deadline	6	4	5	6	3	Y	1

If Y is the minimum value such that Job 6 get scheduled and X is the maximum value possible so that Job 4 will never get scheduled, then the value of $2^{\mathrm{X}+\mathrm{Y}}$ is _____

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the task scheduling problem where the total profit earned is 60. How many non-zero and non-negative values are possible for X such that X is not a part of the profit earned?

Tasks	1	2	3	4	5	6	7	8
Profit	11	16	6	X	5	11	5	6
Deadline	4	5	2	5	2	6	4	7

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the time table at the railway station with arrival and departure times given below.

Rail	Arrival	Departure
Rail A	0700 hrs	0730 hrs
Rail B	0715 hrs	1100 hrs
Rail C	0830 hrs	0900 hrs
Rail D	0845 hrs	0945 hrs

How many the minimum number of platforms are needed so that all the trains can be accommodated as per their schedule?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the recurrence relation given below.

$$T(n) = 2T(n/4) + n^{\alpha}$$

If the time complexity of given relation is $\theta\left(n^{\alpha}\right)$, where $0 < \alpha < 1$, then what is the non-zero minimum value possible for α?(corrected to decimal places)

Your Answer: | Correct Answer: 0.51 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a weighted, undirected graph with positive edge weights. Let (u, v) be an edge in the graph. The shortest path from the source vertex s to u has weight 45 and the shortest path from s to v has weight 60. Which of the following statements is always TRUE?

A. weight $(u, v) < 15$
B. weight $(u, v) > 15$
C. weight $(u, v) \geq 15$
D. weight $(u, v) \leq 15$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

An $m \times n$ matrix is called as Young tableaus (Y), if it satisfies the following properties:

1. Entries of each row are in sorted order from left to right.
2. Entries of each column are in sorted order from top to bottom.
3. Entries may be ∞, which is treated as non-existent elements.

Consider a 4×4 Young Tableau(Y) given below.

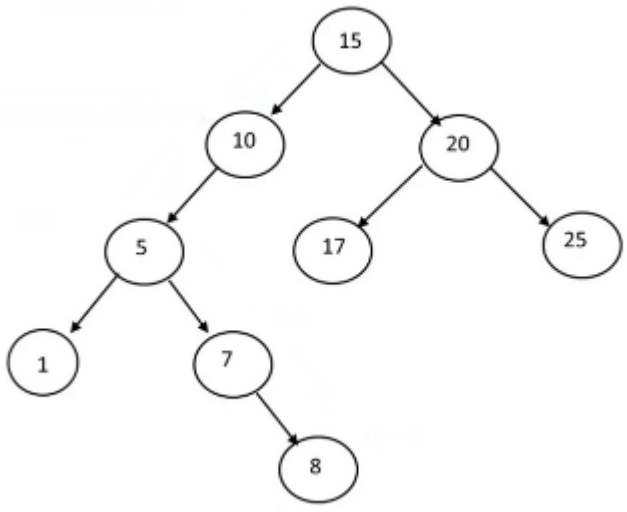
$$\begin{array}{cccc} 3 & 4 & 7 & \infty \\ 5 & x & 8 & \infty \\ y & 15 & \infty & \infty \\ 11 & \infty & \infty & \infty \end{array}$$

If all the entries of integer values in Y is distinct, then which of the following pair of $\langle x, y \rangle$ can be possible?

A. $6, 9$
B. $6, 10$
C. $9, 13$
D. $9, 11$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures



What will be the inorder successor of 10?

A. 1
B. 15
C. 8
D. Does not exist or it is NULL

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider two binary min heaps containing $(2^{\mathrm{L}} - 1)$ and $(2^{\mathrm{R}} - 1)$ nodes. We want to build a new min heap containing all elements of both heaps. It is known that either $\mathrm{L} = \mathrm{R}$ or $|\mathrm{L} - \mathrm{R}| = 1$.

Do not count the time to copy two heap elements into a single array.

A. The most efficient algorithm will take $\mathrm{O}(\log(n))$ where n is the size of the new heap.
B. The most efficient algorithm will take $\mathrm{O}(n)$ where n is the size of the new heap.
C. The most efficient algorithm will take $\mathrm{O}(\mathrm{L})$ only if $\mathrm{L} = \mathrm{R}$ otherwise it will take $\mathrm{O}(n)$.
D. The most efficient algorithm will take $\mathrm{O}(\mathrm{R})$ only if $|\mathrm{L} - \mathrm{R}| = 1$, otherwise it will take $\mathrm{O}(n)$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Let Q be a Queue. Enqueue and Dequeue are usual Queue operations to add and delete element respectively. Q.enqueue(x) adds an element x to the queue. Q.dequeue() performs a dequeue operation on the queue and returns the value that gets dequeued.
Consider the following segment of code:

```
    Q.enqueue(1);
    int count=1;
    do {
        count=count+1;
5.      x = Q.dequeue();
        Q.enqueue( 2*x );
        Q.enqueue( 4*x );
    } while(x != 32);
```

What will be the value of the variable count, when the above segment of code completes its execution?

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Multiple Choice Type Award: 2 Penalty: 0.67 Data Structures

$9, 27, 50, 15, 2, 21,$ and 36 are inserted in an AVL tree. What will be the root of the tree after all insertions?

A. 27
B. 9
C. 15
D. 21

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #42 Multiple Choice Type Award: 2 Penalty: 0.67 Data Structures

Which of the following(s) is/are TRUE-

A. A complete binary tree satisfies the AVL balance condition at each node
B. A full binary tree satisfies the AVL balance condition at each node
C. Insertion in AVL tree requires $\mathrm{O}(\log n)$ rotations
D. Deletion in AVL tree requires $\mathrm{O}(1)$ rotations

Your Answer: Correct Answer: A;C Not Attempted Time taken: 00min 00sec Discuss

Q #43 Numerical Type Award: 2 Penalty: 0 Data Structures

Let a be minimum number of nodes in AVL tree of height 6 and b be maximum number of nodes in AVL tree of height 6 then value of $a + 10b$ is _____

Your Answer: Correct Answer: 1303 Not Attempted Time taken: 00min 00sec Discuss

Q #44 Multiple Select Type Award: 2 Penalty: 0 Data Structures

Which one of the following(s) is/are true about inserting values in BST?

A. Time complexity to Insert N values into an initially empty Binary Search Tree is $\mathrm{O}\left(\mathrm{N}^2\right)$
B. Time complexity to Insert N values into an initially empty Binary Search Tree is $\mathrm{O}(\mathrm{N}\log\mathrm{N})$
C. Time complexity to Insert a value into a Binary Search Tree of size N, where the value you are inserting is the median value when compared to values currently in the tree is $\mathrm{O}(\log\mathrm{N})$
D. Time complexity to Insert a value into a Binary Search Tree of size N, where the value you are inserting is the median value when compared to values currently in the tree is $\mathrm{O}(\mathrm{N})$

Your Answer: Correct Answer: A;D Not Attempted Time taken: 00min 00sec Discuss

Q #45 Multiple Select Type Award: 2 Penalty: 0 Data Structures

Which of the following is/are False about time complexities of Stack and Queue?

A. Dequeue from a queue containing N elements implemented as a circular array takes $\Theta(\mathrm{N})$ time
B. An employee realises that all of his tasks in the stack (implemented using an array) are not required to be done and he decides to empty up the stack and push a new task. He will take $\Theta(\mathrm{N})$ time for this process. Where N is the number of old tasks the stack contains.

C. Reversing a queue using another queue can be done in $O(N)$ time. (Recursion not allowed)
D. Reversing a queue using another two queues can be done in $O(N)$ time. (Recursion not allowed)

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following function that takes the root of the binary tree. Which of the following is true about this function?

```
int complex(BTNode *r) {
    return (r) ? (r->left || r->right) + complex(r->left) + complex(r->right) : 0;
}
```

A. Computes the number of all nodes in the Binary tree
B. Computes the number of all internal nodes in the Binary tree
C. Computes the number of all leaf nodes in the Binary tree
D. Always returns 0 or 1 depending root is null or not null

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

We consider log to the base of 2. Thus for $\log N$ to be a natural number, N must be a power of 2. Let N be an arbitrary number with $N > 1$. Which one of the following statements is true?

A. A binary tree of height $\log N$ can have exactly $N + 1$ Nodes
B. A binary tree of height $\log N$ can have exactly $\log N$ Nodes
C. A binary tree of height $\log N$ can have exactly $N \log N$ Nodes
D. A binary tree of height $\log N$ can have exactly N^2 Nodes

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following is/are true?
In every case consider there are more than 1 nodes.

A. Max element is always leaf node in AVL
B. Min element is always leaf node in AVL
C. 2nd Max in BST is either max's parent or its left child
D. A tree is BST if and only if its inorder traversal is sorted

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

```
    int findHeight(aNode) {
        if (aNode == null)
        //Line1
        int lefth = findHeight(aNode → left);
5.      int righth = findHeight(aNode → right);
        if (lefth > righth) return lefth + 1;
        else return righth + 1;
    }
```

Which of the following should replace the **Line1** to find the height of the subtree for the given node?

A. return -1
B. return 0
C. return 1
D. return max(findHeight(aNode $\rightarrow$ left), findHeight(aNode $\rightarrow$ right))

Your Answer: | **Correct Answer: A** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

In the process of AVL tree insertion or deletion. Let X be the lowest point of imbalance and Y be the child of X in the direction of imbalance. Which of the following is/are true?

A. It is the case of LL rotation if Balance Factor of X is 2 and Balance Factor of Y is 1
B. It is the case of RR rotation if Balance Factor of X is -2 and Balance Factor of Y is -1
C. It is the case of Right-Left rotation if Balance Factor of X is -2 and Balance Factor of Y is 1
D. It is the case of Left-Right rotation if Balance Factor of X is 2 and Balance Factor of Y is -1

Your Answer: | **Correct Answer: C;D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

k^{th} largest element in AVL tree can be found in - (Mark tightest bound only)

A. $\mathrm{O}(\log n)$
B. $\mathrm{O}(k)$
C. $\mathrm{O}(\log n + k)$
D. $\mathrm{O}(n)$

Your Answer: | **Correct Answer: C** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following is/are FALSE?

A. In an AVL tree, If a pointer to some node is given then finding the next smallest element to the given node takes $\mathrm{O}(1)$ time.
B. In a Binary Search tree, If a pointer to some node is given then finding the next smallest element to the given node takes $\mathrm{O}(1)$ time.
C. In array based implementation of heap, if an index of some node is given then finding the next smallest element to the given node takes $\mathrm{O}(1)$ time.
D. In a Binary Search tree, If a pointer to some node is given then finding the next highest element to the given node takes $\mathrm{O}(1)$ time.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following pseudo-code which takes a stack as input and returns some other stack.

```
bar(stack st) {
    stack tmp; //declaring a stack
    int position=1;
    while (!st.empty()) {
        if (position++ % 2 == 1) tmp.push(st.top());
        st.pop();
    }
    return tmp;
}
```

If Stacks are implemented using an array where the bottom of the stack is the 0th index of an array. Which of the following is true about the above function-(odd or even positions are defined as odd or even array indices)

A. The function returns a stack which contains all odd positioned elements from the input stack.
B. The function returns a stack which contains all even positioned elements from the input stack.
C. The function returns a stack which contains all odd positioned elements in the reversed form of the input stack
D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider the function mystery which takes the head of LinkedList as an argument.

```
int mystery(Node* head)
{
    if (head == NULL) return 0;
    else return head->data - mystery(head->next);
}
```

If the linked list contains elements from 1 to n in the order, where the head is pointing to 1. Which of the following is/are true-

A. It returns -50 if n is 99
B. It returns 50 if n is 99
C. It returns -50 if n is 100
D. It returns 50 if n is 100

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Given a complete binary tree with 6 nodes, What is the probability that it forms a (min or max) heap with 6 distinct numbers?

A. $1/18$
B. $1/36$
C. $1/12$
D. $1/6$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #56 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose that stacks and queues are provided as opaque data types, offering only operations to add elements, to remove elements, and to test for emptiness. Suppose that a programmer wants to count the number of elements in a given stack or queue C, which is currently in some state t, using only one auxiliary stack or queue D. The structures C and D can be used in any way possible based on the methods they offer, but C must be restored to its state t after counting its elements.
Counting elements as described above is possible for which of the following data types?

I. C is a queue and D is a queue.
II. C is a stack and D is a stack.
III. C is a queue and D is a stack.

A. None
B. I and II only
C. I and III only
D. I, II, and III only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #57 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Inorder traversing a tree resulted $\mathrm{E\ A\ C\ K\ F\ H\ D\ B\ G}$; the preorder traversal would return.

A. $\mathrm{F\ A\ E\ K\ C\ D\ B\ H\ G}$
B. $\mathrm{F\ A\ E\ K\ C\ D\ H\ G\ B}$
C. $\mathrm{E\ A\ F\ K\ H\ D\ C\ B\ G}$
D. $\mathrm{F\ E\ A\ K\ D\ C\ H\ B\ G}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #58 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

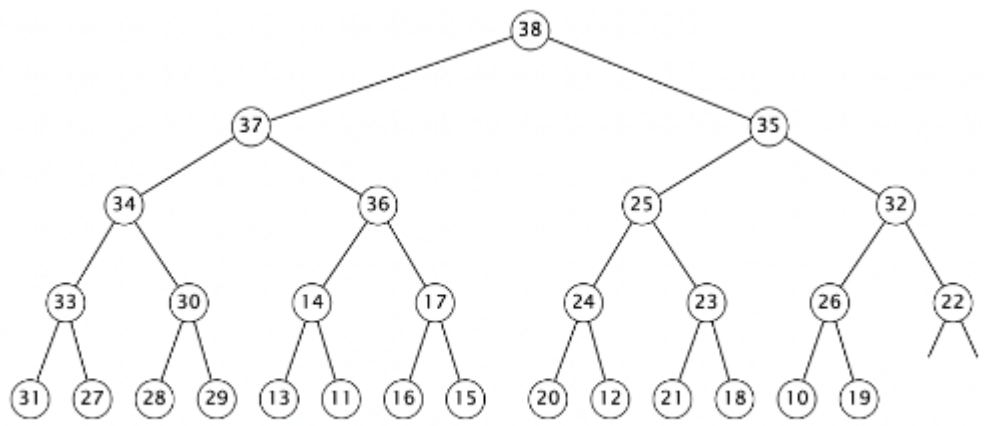
Which of the following is/are true?

A. After hashing n keys into a hash table of size m that uses chaining to handle collisions, we hash two new keys k_1 and k_2. Under the simple uniform hashing assumption, the probability that k_1 and k_2 are hashed into the same table location is exactly $1/m$ with no dependence on the number of keys n.
B. Under the uniform hashing assumption, if we use a hash table of size m with open addressing to hash 3 keys, the probability that the third inserted key needs exactly three probes before being inserted into the table is exactly $\frac{2}{m(m-1)}$.
C. We use a hash table of size m with open addressing to hash n items. Under the uniform hashing assumption, the expected cost to insert another element into the table is at most $1+\alpha$, where $\alpha = n/m$ is the average load.
D. Linear probing is equivalent to double hashing with a secondary hash function of $h_2(k) = 1$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #59 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider the following binary heap.



Suppose that the last operation performed in the binary heap above was inserting the key x. What is/are possible values of x?

A. 19
B. 26
C. 32
D. 10

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #60 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider messages comprised of four symbols - $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$ – each with an associated probability of occurrence: $p(\mathrm{A}), p(\mathrm{B}), p(\mathrm{C}), p(\mathrm{D})$. Suppose $p(\mathrm{A}) \geq p(\mathrm{B}) \geq p(\mathrm{C}) \geq p(\mathrm{D})$.

Which of the following is correct condition (equation or inequality) that is both necessary and sufficient to guarantee that the Huffman algorithm will generate a two-bit encoding for each symbol, i.e., the corresponding Huffman code will actually be a fixed-length encoding using 2 bits for each symbol.

A. $p(\mathrm{C}) + p(\mathrm{D}) > p(\mathrm{A})$
B. $p(\mathrm{C}) + p(\mathrm{D}) > p(B)$
C. $p(\mathrm{C}) + p(\mathrm{D}) \geq p(\mathrm{A})$
D. $p(\mathrm{C}) + p(\mathrm{D}) \geq p(B)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #61 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let T be MST of $\mathrm{G} = (\mathrm{V}, \mathrm{E})$.

Consider two statements $\mathrm{S1}$ and $\mathrm{S2}$.

- $\mathrm{S1}$: If we add a new edge e to G, our T might not be MST anymore. We can obtain a new MST from the old one without recomputing everything in $\mathrm{O}(|\mathrm{V}|)$ time.
- $\mathrm{S2}$: If we add a new node v to G. We can compute a new MST in $\mathrm{O}(\deg(v)|\mathrm{V}|)$.

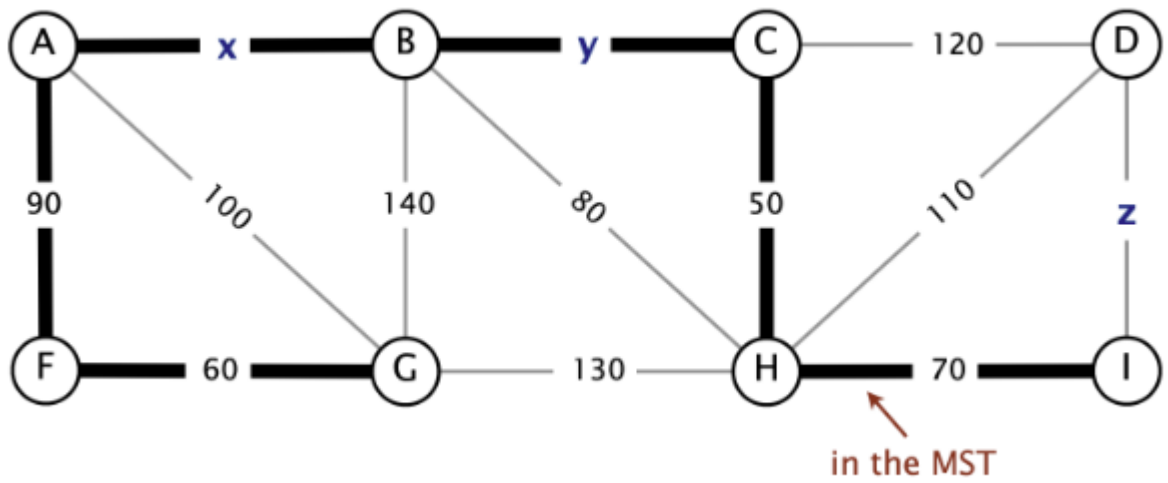
Choose the correct option:

A. $\mathrm{S1}$ is correct but $\mathrm{S2}$ is wrong.
B. $\mathrm{S2}$ is correct but $\mathrm{S1}$ is wrong.
C. Both are correct.
D. Both are false.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #62 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The following diagram shows the set of edges (in thick black lines) selected at some intermediate step of an MST algorithm. Lighter edges are not yet in MST.



Assume that we got above intermediate state using Prim's algorithm.

What is true about the value of z?

A. The minimum value of z must be 50
B. The maximum value of z must be 120
C. The Maximum Value of z must be 140
D. z can take any value

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #63 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Which of the following is correct option about $\mathrm{S1}$ and $\mathrm{S2}$?

- $\mathrm{S1}$: If G is a weighted graph with n vertices and m edges that does *not* contain negative weight cycle, then the iteration of the Bellman-Ford algorithm will reach a fixed point in at most $n-1$ rounds.
- $\mathrm{S2}$: If G is a weighted graph with n vertices and m edges that *does* contain negative-weight cycle, then for *every* vertex v in G the shortest path from v to t in G containing n edges is strictly shorter than the shortest path from v to t in G containing $n-1$ edges.

A. $\mathrm{S1}$ is TRUE but $\mathrm{S2}$ is False
B. $\mathrm{S2}$ is TRUE but $\mathrm{S1}$ is False
C. Both are True
D. Both are False

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #64 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following function that sorts an array of N comparable keys.

```
    void sort(int a[], int N) {
       for (int i = 0; i < N; i++) // outer loop
           for (int j = N-1; j > i; j--)
               if (a[j]swap(a[j], a[j-1]);
5. }
```

Which of these invariants does the above code satisfy at the end of each outer i loop? Mark all that apply:

A. Entries $a[0]$ through $a[i]$ are in sorted order.
B. Entries $a[0]$ through $a[i]$ contain the smallest keys in the entire array.
C. Entries $a[\mathrm{N}-i-1]$ through $a[\mathrm{N}-1]$ are in sorted order.
D. Entries $a[\mathrm{N}-i-1]$ through $a[\mathrm{N}-1]$ contain the largest keys in the entire array.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #65 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Which of the following(s) will evaluate to true

A. $0 \;\&\&\; 0 == 0$
B. $0 \;\&\&\; 1 == 0$
C. $1 \;||\; 0 == 0$
D. $1 \;||\; 1 == 0$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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